

EC-341 Digital System Design

BS (CE)-2k21

Project Proposal Traffic Control System



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Traffic Control System

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Traffic Control System

Abstract:

The traffic control system is designed to optimize the flow of vehicles and enhance road safety through advanced signal management and real-time monitoring. It integrates traffic sensors, cameras, and communication networks to gather data on vehicle movement and road conditions. By utilizing algorithms and artificial intelligence, the system dynamically adjusts traffic signals to reduce congestion and improve traffic efficiency. Additionally, it provides timely information to drivers via variable message signs and mobile applications. This system aims to minimize travel time, reduce fuel consumption, and lower the risk of accidents, contributing to a smarter and more sustainable urban infrastructure.

Introduction:

Our project aims to develop a synchronized digital traffic control system for a four-lane intersection, where each lane is equipped with an independent traffic signal. Our goal is to create an efficient mechanism to manage the flow of vehicles through the intersection. By implementing synchronous control, we aim to enhance traffic flow, minimize congestion, and improve overall intersection safety as well as traffic flow.

Objective:

1.Sequential Progression:

Traffic transitions adhere to a strict sequence

2. System Reset:

The option to reset the system returns traffic to its initial state, halting movement until completed.

3. Lane-Specific Timing:

Each lane receives its designated green light duration.

4. Clearance Periods:

Red light intervals between transitions allow for safe clearance.

5. Timed Delays:

Traffic lights operate with specified delay times for optimal control.

Problem Statement:

Designing an efficient traffic control system for a four-way road interchange presents several challenges that need to be addressed. The primary objective is to develop a state machine capable of managing the flow of vehicles through the intersection while ensuring safety and efficiency. The system must adhere to specific requirements, including sequenced transitions between lanes, reset functionality for returning to the initial state, allocation of movement times for each lane, incorporation of clearance intervals between transitions, and prioritization of main roads for longer green light durations. To address these complexities, a comprehensive understanding of traffic flow dynamics and state machine design principles is essential.

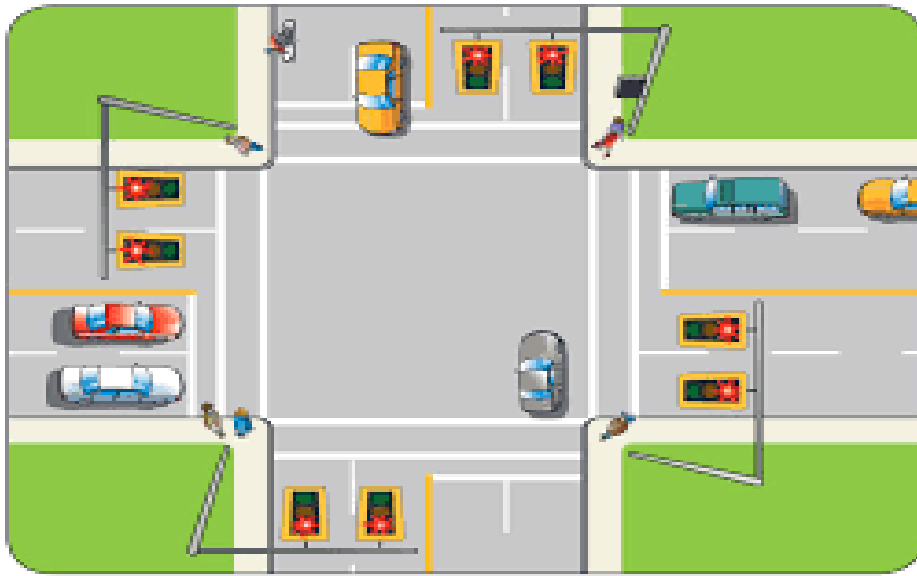
Expected Outcome:

The expected outcomes of the traffic control system include a significant reduction in traffic congestion and improved traffic flow efficiency. It aims to decrease average travel times and fuel consumption, contributing to lower vehicle emissions and environmental impact. Enhanced road safety is anticipated through real-time monitoring and adaptive signal control, leading to fewer accidents. The system also expects to provide better information to drivers, improving their travel experience. Overall, it seeks to create a more sustainable and efficient urban transportation network.

Components Used:

Xilinx Vivado

Project Diagram:



Conclusion:

Designing a state machine for a four-way road interchange enhances road safety and traffic flow. By meeting requirements like sequenced transitions and main road prioritization, the system effectively manages intersections, with ongoing refinement ensuring reliability and improving transportation efficiency for safer road networks in any area.